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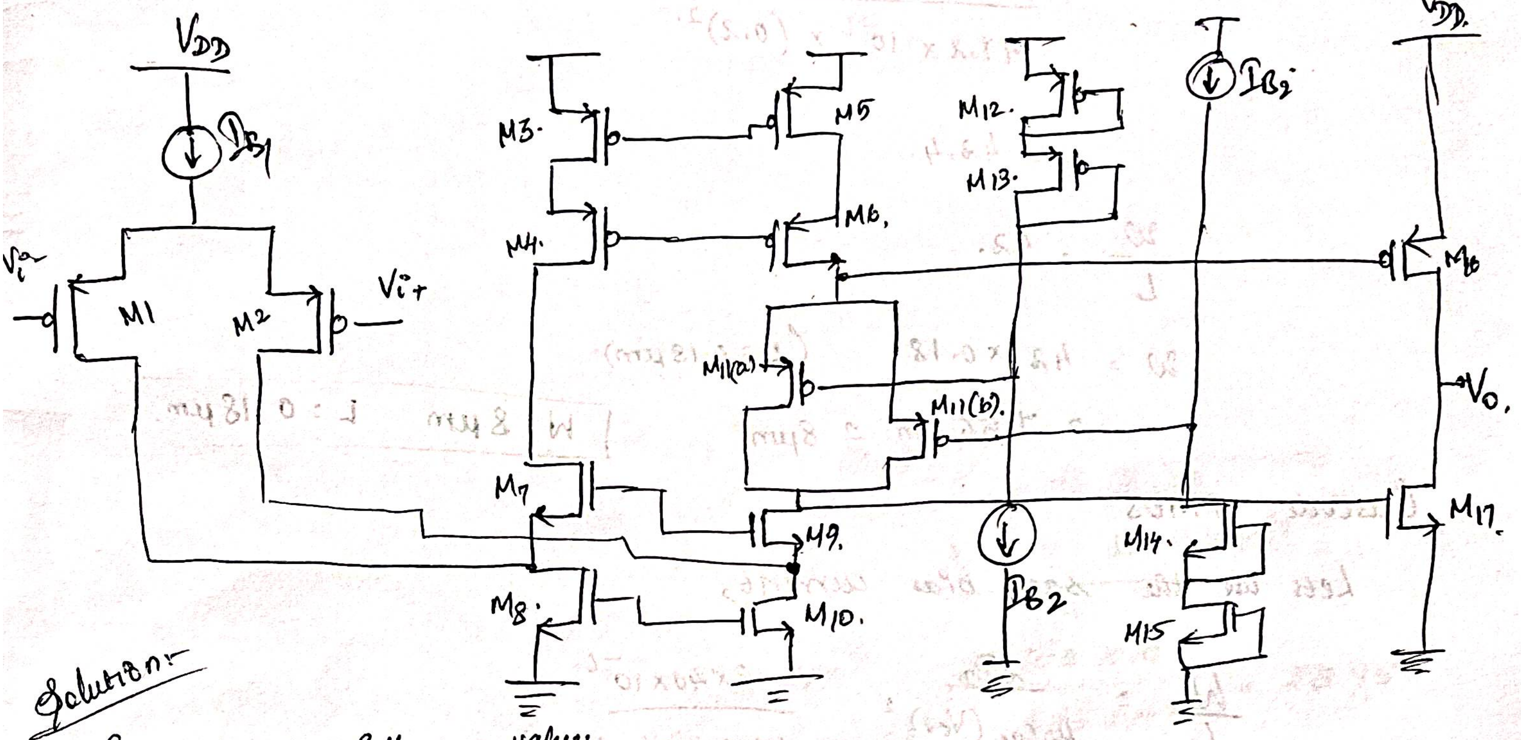
Pre-lab : 5 : Two stage folded cascode Op-Amp with class AB Output stage.



Solution:-

Considering the following values,

$\mu_n C_{ox_n} = 167\ \mu A/V^2$.

$\mu_n C_{ox_p} = 47.2\ \mu A/V^2$.

$V_{ov} = 0.2\,V$. (good choise)

$V_{DD} = 5\,V$.

$I_{B1} = 40\,\mu A$.

$I_{B2} = 40\,\mu A$.

$I_{B3} = 80\,\mu A$.

$C_L = 20\,pF$

$C_C = 5\,pF$

Using $L = 0.18\,\mu m$.

Design Specifications:-

Supply voltage	= 5V
A_{V_0}	> 60dB.
Slew Rate	= 4 V/µs.
GBW	> 10 MHz.
PM	> 45 deg.
Output Swing	= 4.5V.
C_L	= 20pF
Quiescent Power	< 1mW.

$(\mu_n C_{ox}) = 47.2\ \mu$

Input Pair (PMOS) M1, M2.

$$\left(\frac{W}{L}\right) = \frac{2 I_D}{\mu_n C_{ox} (V_{ov})^2}$$

$$= \frac{2 \times 40 \times 10^{-6}}{47.2 \times 10^{-6} \times (0.2)^2}$$

$$= 42.4$$

$$\frac{W}{L} = 42$$

$$W = 42 \times 0.18 \quad (L = 0.18\mu m)$$

$$= 7.56\ \mu m \simeq 8\mu m$$

$\boxed{W = 8\ \mu m \quad L = 0.18\ \mu m}$

Cascode PMOS

Lets use the same bias current,

$$\frac{W}{L} = \frac{2 I_D}{\mu_n C_{ox} (V_{ov})^2} = \frac{2 \times 40 \times 10^{-6}}{47.2 \times 10^{-6} \times (0.2)^2}$$

$$W = 42.4 \times 0.18$$

$$= 7.56\ \mu m$$

$$\simeq 8\ \mu m$$

Calculate $\frac{W}{L}$ for NMOS M7, M9,

$$\left(\frac{W}{L}\right)_{7,9} = \frac{2 I_D}{\mu_n C_{ox} (V_{ov})^2}$$

$$= \frac{2 \times 40 \times 10^{-6}}{167 \times 10^{-6} (0.2)^2} = 12. \times 0.18\ \mu m$$

$$= 2.16\ \mu m$$

$\boxed{W = 2.16\ \mu m}$

NMOS Cascode Sinks (M8, M10).

$$\left(\frac{W}{L}\right)_{8,10} = \frac{2 \times 40 \times 10^{-6}}{167 \times 10^{-6} \times (0.2)^2}$$

$$= 11.976$$

$$\simeq 12$$

$W = 2.16\ \mu m$. when $L = 0.18\ \mu m$

For output devices $I_{-max} = 80\mu A$

M, 16,

$$\left(\frac{W}{L}\right)_{16} = \frac{2 \times 80 \times 10^{-6}}{47.2 \times 10^{-6} \times (0.2)^2}$$

$$= \frac{160 \times 10^{-6}}{1.886 \times 10^{-6}} = 84.8$$

$\boxed{W = 15.26\ \mu m. \quad L = 0.18\ \mu m}$

NMOS, M17,

$$\left(\frac{W}{L}\right)_{17} = \frac{2 \times 80 \times 10^{-6}}{167 \times 10^{-6} \times (0.2)^2} = \frac{160 \times 10^{-6}}{6.68 \times 10^{-6}} = 23.95$$

$\boxed{W = 4.31\mu m, \quad L = 0.18\ \mu m}$

For $M_{11(a)}$ and $M_{11(b)}$ $\quad I_{max} = 80\mu A$.

$$M_{11(a)} \quad \left(\frac{W}{L}\right)_{11} = \frac{2 \times I_{max}}{\mu_n C_{ox} (V_{ov})^2}$$

$$= \frac{2 \times 80 \times 10^{-6}}{167 \times 10^{-6} (0.2)^2} = 23.95$$

$\boxed{W = 4.32\ \mu m \quad L = 0.18\ \mu m}$

$M_{11}(b)$. $\left(\frac{W}{L}\right)_{M11b} = \frac{2I_{max}}{\mu_n C_{ox}(V_{ov})^2}$

$$= \frac{2\times 80\times 10^{-6}}{167\times 10^{-6}\times(0.2)^2}$$

$$= 23.95$$

$$\frac{W}{L} = 24.$$

$\boxed{W = 4.32\mu m}$ $\boxed{L = 0.18\mu m.}$

Slew Rate,

$$SR = \frac{80\mu A}{20pF} = 4V/\mu s.$$

$$Power_{total} \doteq (40+40+80)\mu A \cdot 5V = 0.8mW.$$

For first stage, $g_{m_1} = \frac{2I_{B1}}{V_{ov}} \doteq 0.4mS$

$$r_o = \frac{V_A}{I_{B1}} \simeq 500k\Omega$$

$$A_{V1} \sim 0.4 \cdot 500k\Omega \simeq 200000. > 60dB.$$

$$GBW = 12.7 MHz \text{ for } C_C = 5pF.$$

Output swing, with $V_{GS} \simeq 0.6V$.

↳ easily meets 4.5V swing on 5V rail

Summarizing in a table,

		Function.	W (μm)	L (μm)	I_D
M1, 2	PMOS	Input diff pair	7.56	0.18	40 (tail)
M3–M6.	PMOS	Cascode	7.56	0.18	40
M7–M9	NMOS	Current-mirror	2.16.	0.18	40
M8–M10	NMOS	Folded cascode sink	2.16.	0.18	40
M11a	PMOS	Class AB Control	~~15.56~~ 15.3	0.18	80
M11b.	NMOS	Class AB Control	~~15.3~~ 4.32	0.18	80
M12–15	PMOS/NMOS	Bias & Mirror	15.3μ → PMOS ~~4.32~~ 4 → NMOS	0.18	80
M16	PMOS	Output (pull-up)	4. 15.3	0.18	80
M17.	NMOS	Output (pull-down)	4.32.	0.18	80.

For Transistors → M13, M12,

$I_D = 80\mu$

M12, M13,

$$\left(\frac{W}{L}\right) = \frac{2 \times 80 \times 10^{-6}}{47.2 \times 10^{-6} \times (0.2)^2}$$

$$= 15.26\ \mu m.$$

$I_D = 80\mu$

M14, M15

$$\left(\frac{W}{L}\right) = \frac{2 \times 80 \times 10^{-6}}{167 \times 10^{-6} \times (0.2)^2} = \frac{160 \times 10^{-6}}{6.68 \times 10^{-6}}$$

$$= 23.95$$

$$W = 23.95 \times 0.18$$

$$= 4.31 \mu m$$

Comments on the compensation techniques.

①. Using Miller compensation,

Adding C_c between output of the 1^{st} stage and the output.

② Using feedforward compensation.

→ Capacitor connecting the first stage output to the output using blocking ~~capacitor~~, resistor to introduce a path.

(b) Slew - rate mechanism with equations.

$$SR^{+} = \frac{40\,\mu A}{25\,pF} = 1.6\,V/\mu s.$$

$$SR^{-} = \frac{80\,\mu A}{25\,pF} = 3.2\,V/\mu s.$$

Slew rate is set by available stage currents and total output capacitances